Mi Barrio, My Neighborhood: Barcelona by the Numbers

A Machine Learning Exploration of Tourism and Urban Economics



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Part One — Why Barcelona?

Personal Motivation

Every research project begins with a question that matters personally.

Why this city

Barcelona felt like a natural choice for my semester abroad, partly because of my family's connection to Spain. My father was stationed in Zaragoza while serving in the United States Air Force, and he often spoke fondly of his experiences there, particularly the food, culture, and interactions with everyday people. One story that stayed with me was how his landlord helped him learn Spanish while he was living there. He also shared how respected he felt as a Black resident in Spain. Hearing about these experiences made me interested in experiencing the country for myself.



An overview of the city from the Bunkers del Carmel.

Why this research question

My interest in Barcelona's tourism economy developed before I arrived in Spain. I was already aware of public demonstrations concerning overtourism and the rising cost of housing for local residents. One of my Spanish professors at Vanderbilt had also described Barcelona as increasingly tourist-oriented, particularly because of the growing presence of American visitors compared with other Spanish cities. However, I believed that Barcelona maintained a distinct cultural identity despite its popularity as an international destination. I wanted to investigate how these two realities could coexist by examining tourism patterns, short-term rentals, and their relationship with neighborhood economies.

I was particularly interested in looking beyond the headlines and understanding the economic patterns behind the city's tourism challenges. I wanted to use publicly available data to investigate how tourism activity was distributed throughout Barcelona and whether neighborhoods experiencing greater concentrations of tourist accommodations also experienced differences in housing prices.

What I hoped to learn

Beyond strengthening my technical abilities, I hoped to create something that could be useful to people interested in visiting Barcelona. My intended audience was not limited to American tourists but included international visitors from different linguistic and cultural backgrounds. This influenced my approach to the project, particularly my decision to incorporate multilingual sentiment analysis when examining visitor reviews.

Ultimately, I wanted to create a resource that would help visitors make more informed decisions about where they stay and how they experience Barcelona. My intention was never to discourage tourism, but rather to encourage a more thoughtful approach to visiting the city.

Part Two — Living in the City

Observations and Reflections from the Semester

Data tells one story about my time in Barcelona. Living there tells another. This section connects my personal experience in Barcelona to the themes I later studied.

The neighbourhood I lived in

During my semester abroad, I lived in Aparto Pallars, an international student residence located in the northeastern part of Barcelona. The residence was only a few minutes from the Diagonal Mar shopping center and within walking distance of the beach. Although the area was close to major attractions and shopping, it felt more residential than tourist-oriented. I was right next to an elementary/middle school, and the surrounding streets were filled with everyday activity. There was a Mercadona around the corner, along with smaller shops and local restaurants that became part of my routine.

One of the things I appreciated most about living in this neighborhood was the opportunity to observe everyday life outside of the major tourist areas. On weekdays, I would see schoolchildren heading to class, exchange students going about their day, and residents making their way to the metro. My closest station was Selva de Mar on the yellow L4 line. The train was usually relatively quiet when I boarded, but it gradually became more crowded as it traveled toward the city center. Even something as ordinary as taking the metro gave me a better sense of how different parts of Barcelona connected.



Local shops near Selva De Mar metro station

Noticing tourism in daily life

Although my neighborhood felt relatively residential, tourism was difficult to avoid throughout Barcelona. I noticed a particularly different atmosphere when visiting a friend who lived near Arc de Triomf. Her apartment building even had an Honest Greens restaurant attached to it, and the surrounding area felt much more connected to the city's visitor population.

One of the easiest ways I learned to recognize businesses catering to tourists was through language. Restaurants and businesses that seemed more tourist-oriented often offered English menus, while smaller, more local establishments were more likely to have menus exclusively in Spanish or Catalan. I also became increasingly aware of how recognizable American tourists could be, sometimes before they even spoke. The contrast was especially noticeable in heavily visited areas of the city, such as the area surrounding the Sagrada Família, where the concentration of visitors was much more apparent.



The iconic church, La Sagrada Familia, that attracts tourists worldwide

Conversations and community

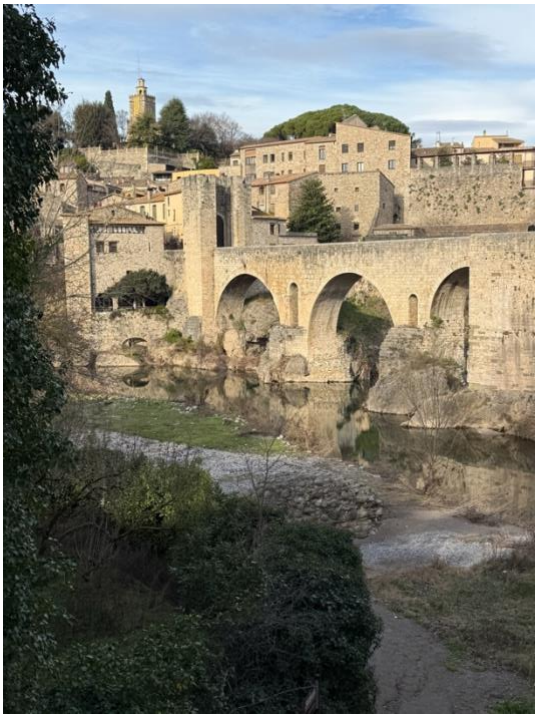
Some of my most meaningful experiences came from ordinary interactions rather than organized excursions. I regularly practiced Spanish when speaking with local shopkeepers, restaurant employees, and other residents. English was primarily the language I used with my American friends, but I made an effort to use Spanish in my everyday interactions. I also made friends at Pompeu Fabra University, which gave me another opportunity to connect with people outside my international student residence. When locals asked where I was from, I occasionally told them, "Soy de Jamaica." Part of the reason was that I did not want to be immediately recognized as an American.

Neighborhoods I explored

Beyond Barcelona, I had the opportunity to visit several places throughout Catalonia. One of my favorite destinations was Tossa de Mar. I especially enjoyed its coastline and the architecture surrounding the castle, which made the town feel distinct from the larger urban environment of Barcelona.

I also visited Besalú, where I was particularly interested in the medieval architecture and surrounding volcanic landscape. I appreciated the balance between locals and visitors who seemed interested in experiencing the town's history and culture. Compared with some of Barcelona's busier tourist areas, Besalú had a much more relaxed atmosphere.

In Tarragona, I visited the Roman amphitheater, where gladiators once fought. I also traveled to Figueres to visit Salvador Dalí's home and museum, where I learned more about his life and surrealist approach to art. Each destination offered a different perspective on the region's history, culture, and relationship with tourism.



Some of the architecture Besalú has to offer

Part Three — The Data

What I Studied and How

This section explains my project's methodology in plain language.

The datasets

The project drew on four primary data sources, all publicly available:

Inside Airbnb

A public dataset of every active Airbnb listing in Barcelona, scraped in 2025. Contained 19,410 listings with prices, locations, room types, ratings, and over one million visitor reviews. This was the backbone of all three analyses.

Open Data BCN — Tourist Housing Register (HUTs)

Barcelona's official register of licensed tourist apartments (habitatges d'us turistic). Contained 10,716 individual registrations with neighborhood locations and bed counts. This was used as the primary indicator of formal tourist housing pressure in each neighborhood.

Open Data BCN — Population and Density

Official Barcelona census data providing population counts and geographic area for all 73 neighborhoods. Used to normalize Airbnb and HUT counts into per-capita rates, enabling fair comparisons between large and small neighborhoods.

Barcelona Neighborhood Boundaries (GeoJSON)

Official polygon boundaries for all 73 Barcelona neighborhoods, used to draw the choropleth maps in the interactive dashboard.

Three analytical methods

The project applied three distinct machine learning and data analysis techniques, each answering a different research question.

Regression Analysis — Predicting Price

A Random Forest regression model was trained on 11,725 Airbnb listings to predict nightly price from 17 features — including apartment size, room type, review scores, host characteristics, geographic location, and neighborhood tourist housing density. The model achieved an R^2 of 0.614, meaning it explained 61.4% of price variation from publicly available data alone.

Clustering Analysis — Neighborhood Types

K-Means clustering grouped Barcelona's 71 neighborhoods into four distinct types based on price level, Airbnb density, tourist apartment density, population density, rating quality, and listing availability. The algorithm found the groupings independently — without being told in advance what the neighborhood types should be.

Sentiment Analysis — Visitor Reviews

A multilingual sentiment model from Hugging Face Transformers was used to analyze the emotional tone of a stratified sample of 29,753 Airbnb visitor reviews, with up to 500 reviews sampled per neighborhood from a corpus of more than one million reviews dating from 2015 to 2025. Each review received a compound sentiment score from -1 (most negative) to +1 (most positive).

Technical tools

All analysis was conducted in Python using Jupyter Notebooks. Key libraries: pandas and numpy for data processing, scikit-learn for machine learning, matplotlib and seaborn for scharts, Plotly and ipywidgets for the interactive dashboard, Hugging Face multilingual sentiment analysis for natural language processing.

Part Four — Key Findings

What the Data Revealed

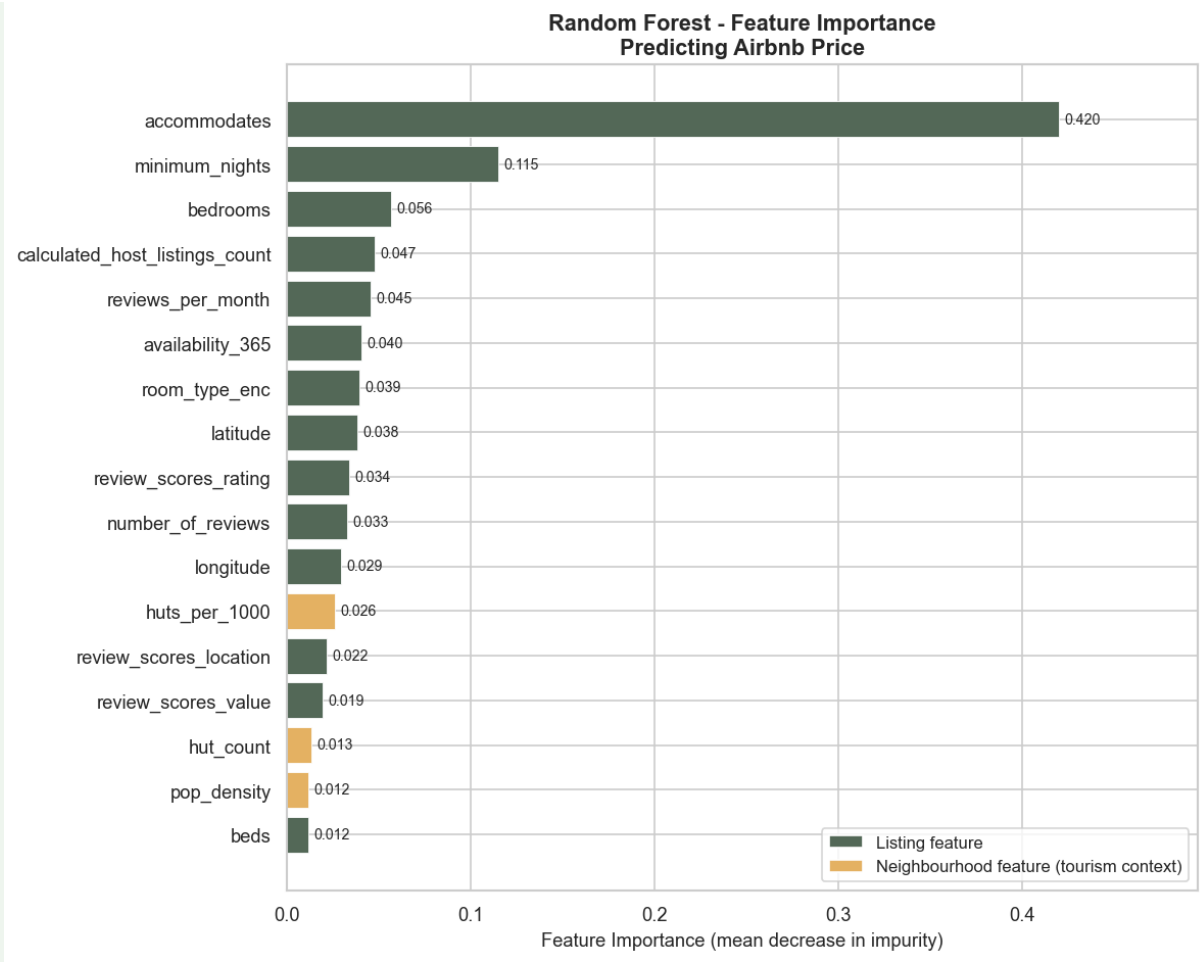
This section presents the most significant results from each analysis

Finding 1: Price is driven by size, not location

The most important predictor of an Airbnb listing's nightly price is how many guests it accommodates — responsible for 42% of the model's predictive power. Room type (entire home vs. private room) and number of bedrooms follow. Neighborhood characteristics, including tourist housing density, matter but are secondary to the apartment's own physical features.

Entire home median price: €168/night *vs. €62/night for a private room — a 2.7x gap*

Model accuracy (R^2): 0.614 *— explains 61% of why prices vary*



Feature importance rankings from the Random Forest model. Bar length indicates each feature's contribution to price predictions. Neighborhood tourism features are shown in gold.

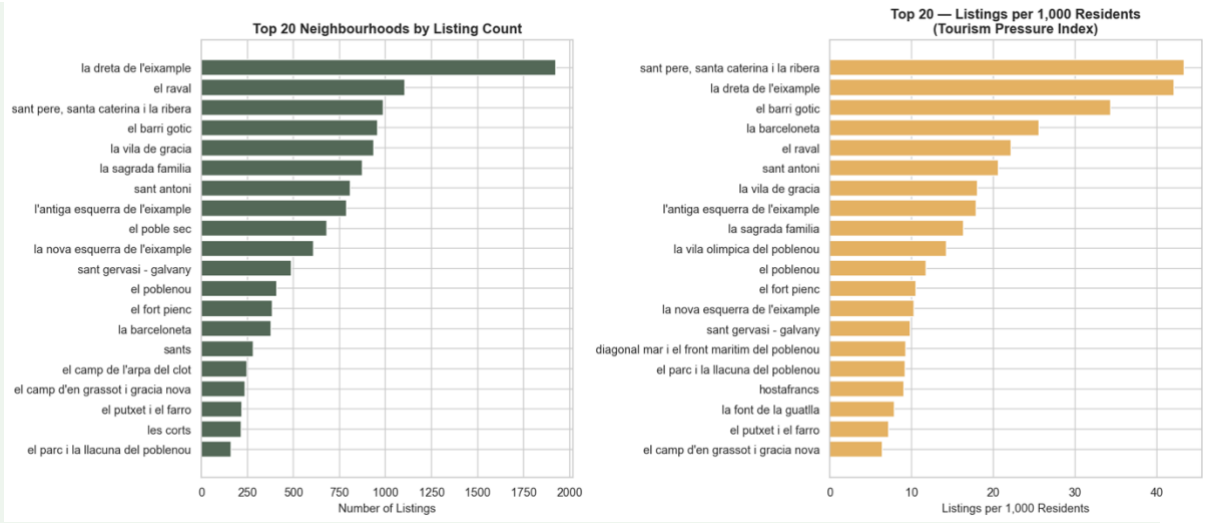
Finding 2: Tourism is concentrated, not spread evenly

Airbnb activity and licensed tourist apartments are not distributed uniformly across Barcelona's 71 neighborhoods. A small number of central barris account for the vast majority of tourist housing activity.

la Dreta de l'Eixample: 1,923 Airbnb listings *— more than double the second-highest neighborhood*

Sant Pere & la Ribera: 43 listings per 1,000 residents *— roughly 1 tourist apartment for every 23 residents*

la Dreta de l'Eixample HUTs: 1,802 licensed tourist apartments *with 12,489 tourist beds*



Left: raw listing counts by neighborhood. Right: listings per 1,000 residents, revealing which neighborhoods are most saturated relative to their residential population.

Finding 3: Four distinct neighborhood types emerge

The clustering analysis identified four neighborhood types that reflect real geographic and economic patterns in Barcelona.

High-Tourism / High-Price (18 neighborhoods)

The tourist core. Includes Barri Gòtic, el Raval, Eixample sub-districts, la Vila de Gràcia, la Barceloneta. Median price €144/night, 18x the Airbnb density of peripheral neighborhoods, and 12.5 licensed tourist apartments per 1,000 residents.

Mid-Price Mixed (36 neighbourhoods)

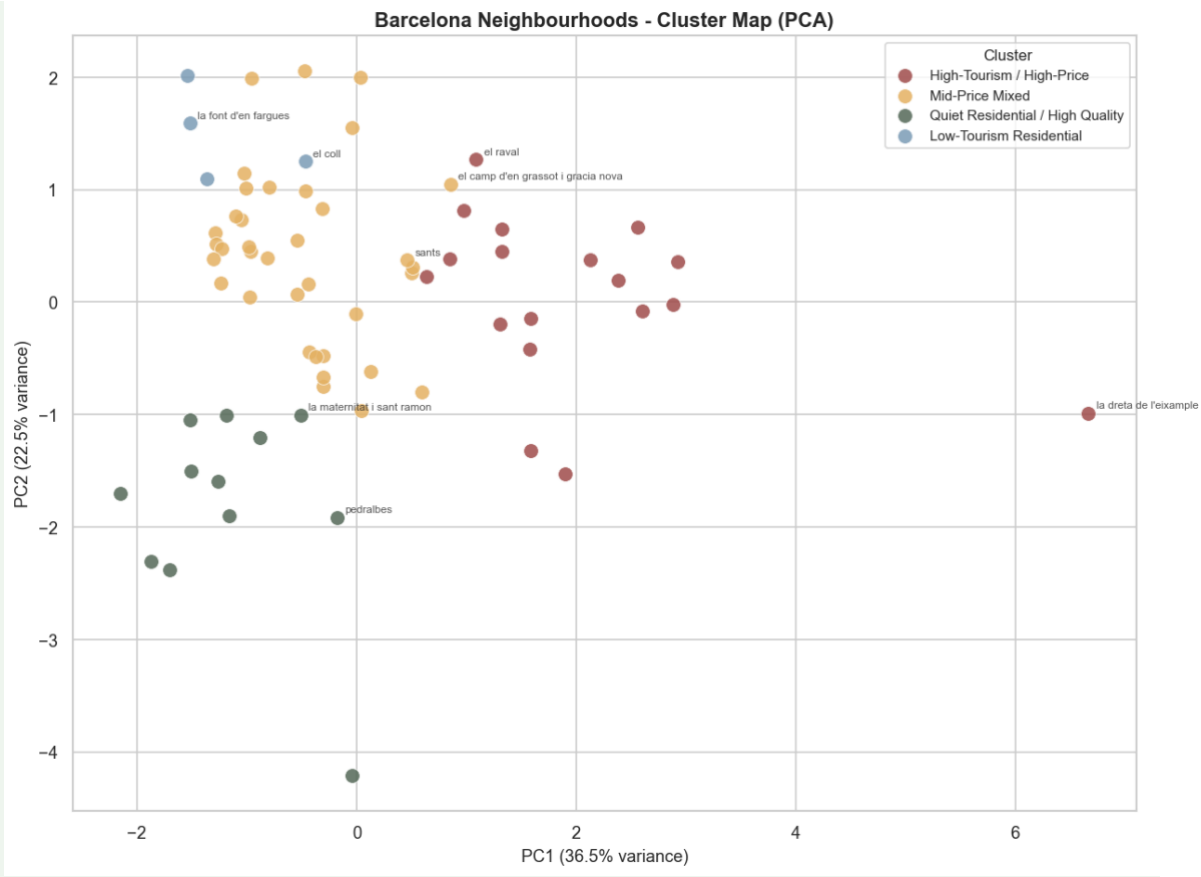
The largest cluster at 51% of all neighbourhoods. Barcelona's broader residential fabric — moderate tourism, prices around €95/night, a mix of locals and some visitors.

Quiet Residential / High Quality (12 neighborhoods)

Low tourism intensity but the highest average ratings (4.8) and most availability. Vallvidrera, Pedralbes, la Maternitat. Listings here are carefully managed in residential settings.

Low-Tourism Residential (4 neighbourhoods)

The most peripheral barris — el Coll, la Vall d'Hebron, la Font d'en Fargues, la Trinitat Vella. Minimal Airbnb presence, lowest prices at €58/night median.



Each dot is one of Barcelona's 71 neighborhoods, positioned by its tourism and economic profile. Color indicates cluster membership. Dots close together are similar neighborhoods; dots far apart are different.

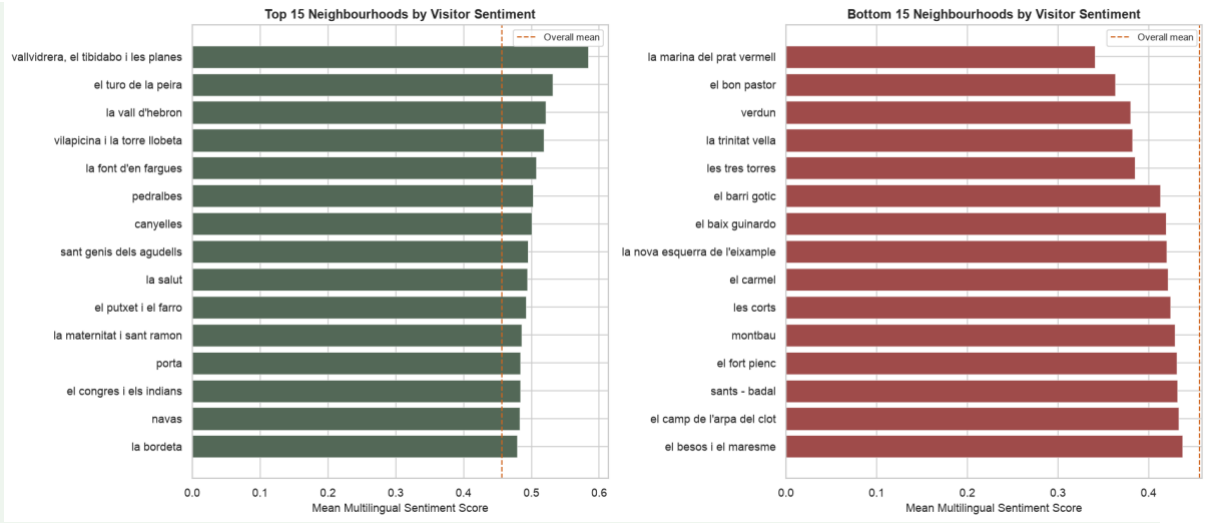
Finding 4: Visitors are most satisfied in the tourist core

The sentiment analysis produced an interesting finding: visitors report positive experiences across Barcelona, with some of the highest sentiment found in lower-tourism residential neighborhoods rather than the most touristically saturated areas.

Overall sentiment: **0.433 mean compound score** — *88% of reviews are positive*

Low-Tourism cluster: **0.481 mean sentiment** — *highest of all four cluster types*

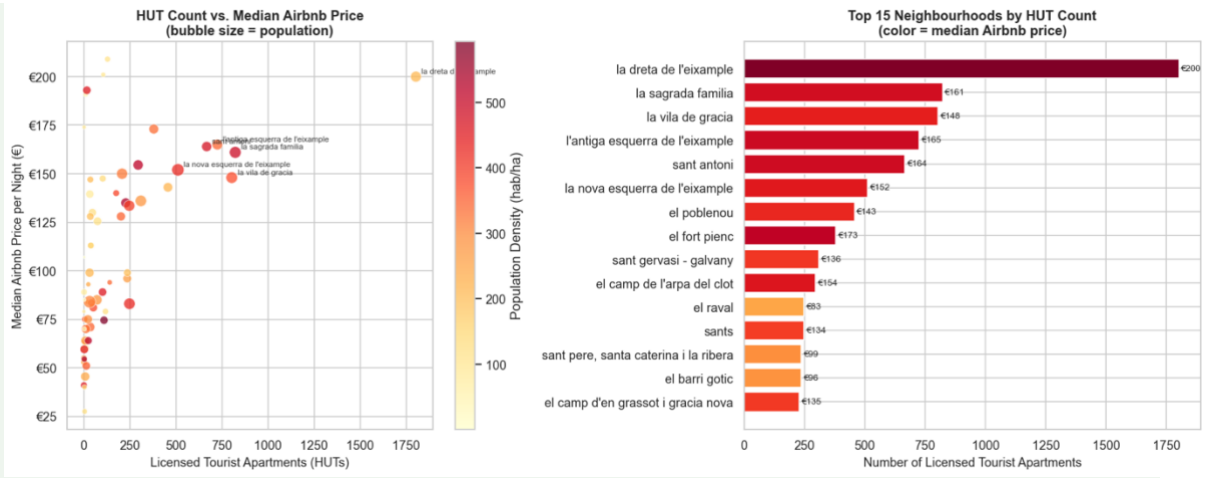
Vallvidrera/Tibidabo: **0.619** — *highest individual neighborhood score*



Top and bottom 15 neighborhoods by mean visitor sentiment score. Scores above 0 are positive; the orange dashed line marks the city average.

Finding 5: Neighbourhood tourist housing density predicts price

Across all three analyses, a consistent signal emerged: neighbourhoods with more licensed tourist apartments per resident tend to have higher Airbnb prices. This was confirmed by the exploratory correlation (r = 0.526), the regression model (huts_per_1000 ranked 12th of 17 features), and the clustering results (the high-tourism cluster has 31x more HUTs per resident than the quiet residential cluster).



Relationship between licensed tourist apartment density and median Airbnb price by neighborhood. Bubble size represents population; color indicates population density.

Part Five — Connecting Data to Experience

What Living in Barcelona Taught Me That the Data Could Not

The numbers describe patterns. Living in the city explains why those patterns exist and what they feel like from the inside.

When the data confirmed what I observed

Working with the data allowed me to look back on my time in Barcelona from a different perspective. Some findings aligned with what I had observed, while others revealed patterns I had not considered at the time. For example, although I was familiar with La Dreta de l'Eixample because my study abroad center was located nearby, I had never really thought about how heavily concentrated Airbnb listings and tourist accommodations were in that neighborhood. My own living situation also gave me a different perspective from someone visiting Barcelona for a short vacation. Living near Selva de Mar in a relatively residential neighborhood allowed me to become familiar the rhythm of everyday life. Rather than simply visiting Barcelona, I felt as though I was temporarily living there. Barcelona became a second home to me instead of just a vacation destination.



People gather at the Cathedral of Barcelona in the heart of the Gothic Quarter

When the data surprised me

One thing I had not fully considered before conducting this project was the extent to which tourism accommodations were concentrated in certain neighborhoods. Although I recognized that areas with major attractions would naturally draw more visitors, I had not thought much about the number of Airbnb listings and licensed tourist apartments located near places I regularly visited. I was also surprised that the physical characteristics of an Airbnb listing, particularly its guest capacity, were more influential in predicting price than some of the surrounding neighborhood characteristics. I initially assumed that location would have a greater impact, especially in a city where certain areas are particularly popular with tourists. These findings challenged some of my initial assumptions.

The human side of the numbers

Before beginning this project, I was already aware of the conversations surrounding overtourism, housing affordability, and protests in Barcelona. However, working with the data made these issues feel more concrete and helped me better understand the perspectives of local residents. I came to recognize that frustration surrounding overtourism is not necessarily about disliking Americans or tourists; for many residents, it is about being able to afford to live in the place they call home. I still believe tourism brings valuable economic opportunities, and I enjoy supporting local businesses and approaching travel as an opportunity to learn about different cultures. Ultimately, I believe there needs to be a happy medium between encouraging tourism and protecting residents' quality of life. Tourism should benefit a city without overshadowing housing affordability concerns or the needs of the people who call that city home.



Many visitors seen on a rainy day at a residential garden known as Park Güell

Barcelona and global tourism economics

Barcelona's experience also reflects broader questions surrounding tourism and urban economics that cities around the world face. Tourism can generate significant economic activity by supporting local businesses, creating employment opportunities, and attracting international visitors. However, managing that growth requires balancing the interests of visitors, businesses, and permanent residents. Barcelona's approach to regulating tourist accommodations illustrates how cities may respond when tourism intersects with housing affordability and the availability of residential properties. This project helped me recognize that these challenges are not unique to Barcelona, but part of a larger conversation about how popular destinations can benefit from tourism while preserving the communities that make them attractive in the first place.

Part Six — Conclusions

What I Learned and What Comes Next

Answers to the original research questions

The project set out to answer three questions. Here is what the analysis found:

How are Airbnb prices distributed across Barcelona?

Prices cluster between €69 and €212/night for most listings, with a median of €130. Entire homes cost 2.7x more than private rooms. The distribution has a long right tail — most listings are affordable but a small number of premium properties pull averages upward. Price is driven primarily by apartment size, not neighbourhood prestige.

Which neighbourhoods are most affected by tourism?

Tourism is highly concentrated. Sant Pere & la Ribera (43 listings per 1,000 residents), la Dreta de l'Eixample (42), and el Barri Gòtic (34) face by far the highest Airbnb saturation relative to their populations. La Dreta de l'Eixample alone has 1,802 licensed tourist apartments — nearly a fifth of all HUTs in the city.

Does tourist housing density correlate with higher prices?

Yes, consistently across all three analyses. The correlation between HUT density and median price is r = 0.526. The regression model identifies huts_per_1000 as a meaningful predictor. The High-Tourism cluster — with the most HUTs — charges the highest prices. The relationship is real but not deterministic: apartment size still matters more than neighbourhood.

Learning goals — how they were met

One of my primary goals for this project was to strengthen my data analysis skills while applying concepts I had learned in the classroom to a real-world issue. I am particularly proud of combining several datasets, building multiple machine learning models, conducting sentiment analysis, and creating visualizations to communicate my findings. I also found a creative way to document my study abroad experience by integrating my academic interests in Economics, Public Policy, Data Science, and Spanish for the Professions. Economics provided a framework for examining tourism and housing markets, Public Policy helped me consider Barcelona's responses to these issues, Data Science provided the tools to conduct the analysis, and Spanish helped me engage with the cultural and linguistic environment in which the research took place.

What I would do differently

If I were to approach this project again, I would place greater emphasis on planning the research process and establishing a clear timeline for each stage. Working with multiple datasets, cleaning inconsistent neighborhood names, and troubleshooting

analytical methods required more time than I initially anticipated. I would also explore additional ways to incorporate the perspectives of Barcelona residents. While Airbnb listings and visitor reviews provided useful insight into tourism patterns, they cannot fully capture how residents experience changes in their neighborhoods or the pressures they face in the housing market. Incorporating resident perspectives or additional housing data could provide a more comprehensive understanding of the relationship between tourism and local communities.

Final reflection

One of my biggest takeaways from this project is that there is not always a perfectly neat way to capture complex urban dynamics, but that does not mean we should not try to understand them. Combining quantitative data with my own experiences helped me look at Barcelona beyond its identity as a popular tourist destination and consider the different perspectives of the people who visit and those who call it home. My ultimate professional goal is to pursue real estate law, and this project gave me another opportunity to examine housing dynamics from a critical perspective. It reinforced the importance of understanding issues from multiple perspectives, recognizing the limitations of available evidence, and finding meaningful ways to communicate research. I believe these skills will provide a valuable foundation for future research in law school and throughout my professional career.



A farewell to the beautiful Barcelona!